



Musa Buğra Demir

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EDUCATION AND TRAINING

30/08/2023 – CURRENT İzmir, Türkiye
COMPUTER ENGINEERING Ege University

Website <https://ege.edu.tr> | **Level in EQF** EQF level 6

LANGUAGE SKILLS

Mother tongue(s): **TURKISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	B2	B1	B2
SPANISH	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Skills & Competencies

Programming Languages: C, C++, Python, C#, Java, SQL, PyPy3, Dart, Flutter, Solidity

AI & Computer Vision: OpenCV, YOLO, PyTorch, TensorRT, Image Processing Pipelines

Robotics & Simulation: ROS 2, Gazebo Harmonic, VRX, Rviz2, Autonomous Navigation Systems

Game Development & 3D Design: Unity, Unreal Engine, Blender, SolidWorks

Hardware & Systems: Arduino, STM32, ESP32, Signal Processing, Telemetry Data Flow, Microcontrollers

Blockchain Technology: Smart Contracts, Web3 Concepts, Monad and Sui Environments

Design & Multimedia: Adobe Photoshop, Canva

Soft Skills & Leadership: Cross-functional Team Leadership, Event Coordination, Crisis Management, Public Speaking, High-Impact Presentations, Proactive Ownership, Professional Reporting, Algorithmic Thinking, Multidisciplinary Collaboration, STEM Mentoring

MANAGEMENT AND LEADERSHIP SKILLS

Leadership & Event Management Experience

Head of Organization & Coordination Teams | İş Hayatında Kelebek Etkisi'9 (IEEE Ege WIE) – March 2026 Directed cross-functional teams for a high-impact social awareness event. Managed end-to-end logistics, mitigated real-time crises, and led the team to deliver a seamless event experience.

Head of Coordination Team | SciTech Conference (IEEE Ege EMBS) – February 2026 Spearheaded the coordination efforts for a comprehensive biomedical conference. Fostered strong teamwork and agile problem-solving to ensure smooth operational flow and participant satisfaction.

Coordination Team Member | GreenInvest (EU-Supported) – March 2026 Facilitated successful event execution for a major EU-backed initiative. Collaborated with a diverse, international team and managed global communications in English to overcome logistical challenges.

Coordination Team Member | CyberEge24 (IEEE Ege CS) – May 2024 Played an active role in hosting Izmir’s largest technology and cybersecurity event. Contributed to large-scale crowd management, dynamic team coordination, and on-the-spot crisis resolution.

Coordination Team Member | IEEE Aegean Regional Meeting – November 2023 Supported the successful execution of a major regional gathering at Ege University. Developed foundational skills in event operations and adaptive teamwork.

Active Participant & Networker Continuously expanding cross-disciplinary knowledge and professional adaptability through active engagement in diverse summits and camps, including EMBCAMP, WIELAND, Izmir Energy Summit, Etkileşim & Gelişim Kampı, CSCON, and ROBOLEAGUE.

28/06/2025 – CURRENT

IEEE Ege University Student Branch Treasurer of the Board

In 2025, I was elected Treasurer of the IEEE Ege University Student Branch Board. **My responsibilities** in this role have included **financial planning, budgeting, and reporting** for the branch. I collaborated with the Sponsorship and Event Teams to secure funding, **allocate resources effectively**, and **ensure the financial sustainability** of our projects. This position has **fortified my aptitudes** in **financial administration, organizational strategy**, and **interdisciplinary teamwork**.

12/07/2025 – CURRENT

Board Member Responsible for IEEE Computer Society, IEEE Communications Society, IEEE Women in Engineering Society

As the **board member** responsible for the **IEEE Computer Society, IEEE Woman in Engineering Society** and the **IEEE Communications Society**, and as the **chair of the Communications Society**, I provided **strategic leadership for interdisciplinary** activities and ensured the societies were **effectively coordinated**. I **planned** and executed **seminars, technical workshops**, and outreach initiatives that fostered innovation and high member engagement. I also **oversaw** committee operations, **resolved organizational and logistical challenges**, and **strengthened collaboration** by facilitating **clear communication between teams**. Through these efforts, **I improved the branch’s operational efficiency and enhanced its overall impact**.

WORK EXPERIENCE

IEEE EGE STUDENT BRANCH | TECHNICAL TEAM MEMBER – 10/10/2025 – Current

Ege Mavi İnci USV (Unmanned Surface Vehicle) Team

- **Roles:** Image Processing, Communication Algorithms, Simulation Testing, Mechanical Design, Autonomous System Development
- Developed and integrated computer vision algorithms using **OpenCV** and **YOLO** to enable robust autonomous navigation and environmental perception.
- Designed and optimized communication algorithms in **Python** and **C++** for secure and reliable data transmission between the vehicle and ground control.
- Conducted extensive simulation tests utilizing **ROS 2** and **VRX** to validate system performance and safety prior to real-world deployment.
- Contributed to the mechanical design and structural integrity of the vehicle using **SolidWorks**, ensuring optimal hydrodynamic performance.
- Spearheaded the overall preparation and integration of the autonomous system architecture.
- **Key Technologies:** C++, Python, ROS 2, VRX, OpenCV, YOLO, SolidWorks

Ege ODBARS UGV (Unmanned Ground Vehicle) Team

- **Roles:** Image Processing, Communication Algorithms & Systems, Simulation Testing, Mechanical Design, Autonomous System Development
- Engineered real-time image processing pipelines utilizing **OpenCV** and **PyTorch** for dynamic obstacle detection, tracking, and path planning.
- Managed comprehensive wireless communication systems and telemetry networks to guarantee low-latency remote operation.
- Executed rigorous simulation testing within **Gazebo Harmonic** and **Rviz2** to verify the reliability of both software and hardware components under various conditions.
- Assisted in the mechanical modeling and physical assembly of the ground vehicle.
- Played a key role in developing and fine-tuning the vehicle’s core autonomous capabilities using **ROS 2**.
- **Key Technologies:** C++, Python, ROS 2, Gazebo Harmonic, Rviz2, OpenCV, PyTorch

Ege CubeSat Team

- **Roles:** Communication Algorithms & Systems, Simulation Testing, Mechanical Design

- Designed and implemented reliable satellite communication links and sophisticated signal processing algorithms using **C** and **C++**.
- Performed advanced simulation tests to ensure system resilience in simulated space environments.
- Participated in the mechanical design and structural analysis of the satellite via **SolidWorks**, adhering to strict aerospace durability standards.
- **Key Technologies:** C, C++, Python, SolidWorks

Ege Çelikgöz Çelikkubbe Air Defense Systems Team

- **Roles:** Image Processing, Communication Algorithms & Systems, Simulation Testing, Mechanical Design, Autonomous System Development, Weapon Systems Design
- Implemented advanced image processing models using **TensorRT** and **YOLO** for highly accurate, real-time target identification and tracking in airspace.
- Developed resilient communication systems and algorithms in **C++** to maintain secure, uninterrupted command and control links.
- Validated complex defense scenarios and system responses through comprehensive simulation environments, leveraging tools like **Unreal Engine**.
- Contributed to the mechanical design of the defense platform and weapon systems using **SolidWorks**.
- Integrated multiple subsystems via **ROS 2** to prepare a fully functional, autonomous air defense architecture.
- **Key Technologies:** C++, Python, ROS 2, TensorRT, YOLO, Unreal Engine, SolidWorks

AEGEAN CRAFTERS | SOFTWARE DEVELOPER & LEVEL DESIGNER – 07/09/2025 – Current

- Developing core game mechanics and systems using **C# and Unity engine**.
- Designing immersive and balanced game environments as a **Level Designer**, focusing on player flow and spatial aesthetics.
- Collaborating with a multi-disciplinary team to ship high-quality game projects within strict deadlines.

● COMPETITIVE PROGRAMMING

Hackathons

Competitor | IEEEExtreme Global Programming Competition

- Competed in a rigorous 24-hour global algorithmic programming challenge, **ranking 33rd in Turkey and 1071st globally**.
- Applied advanced data structures, dynamic programming techniques, and rapid problem-solving skills to tackle complex technical problems under strict time constraints.
- **Key Technologies:** C#, Python, Java, PyPy3

Award Winner | Monad Blitz Hackathon

- Engaged in a fast-paced development environment and **won the "Best Idea" and "Most Sustainable Idea" categories**.
- Demonstrated technical adaptability and agile teamwork to deliver functional, innovative prototypes within limited timeframes.
- **Key Technologies:** Smart Contracts, Monad Testnet, Solidity, Dart, Flutter, GridSingularity, Python, Docker, Forge

● VOLUNTEERING

STEM & Social Impact Volunteer | IEEE Ege Student Branch

Actively engaged in multiple social responsibility initiatives to promote engineering, empower youth, and foster a diverse technological community.

- **Children in Power:** Led hands-on workshops and maker culture activities, guiding **over 200 students** in exploring basic technology, robotics, and creative problem-solving.
- **Hour of Code:** Mentored **more than 150 young learners** in foundational coding principles, fostering an early interest in computer science and software development.
- **Teach and Make:** Facilitated interactive educational sessions, teaching practical engineering concepts and hands-on skills to **over 100 students**.
- **Akademiden Liseye: Kadın Mühendisler:** Coordinated efforts to connect female high school students with successful academics and professionals, promoting gender diversity and inspiring the next generation of women in STEM.
- **STAR Project (Student-Teacher and Researcher):** Facilitated educational sessions to introduce core engineering concepts to younger students, bridging the gap between academic research and early education.